

# Supplementary material: Masking cannot remove what is inside the region of interest

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## S1. Pre-registration index

Every experiment was specified in a dated document committed to the public repository before any model was fitted on the corresponding data; amendments were committed before the amended analysis. Git history provides the time stamps.

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| Document                         | Scope                                                                          |
|----------------------------------|--------------------------------------------------------------------------------|
| PREREGISTRATION_ISIC2019_TRAPS   | dermoscopy hair traps (and the author's E13 specification)                     |
| PREREGISTRATION_THYROID_TRAPS    | thyroid calipers: real traps, controlled sweeps (DINOv2, MedSigLIP)            |
| PREREGISTRATION_OVARY_TRAPS      | ovarian calipers: real traps, controlled sweep                                 |
| PREREGISTRATION_CAPSULE_TRAPS    | capsule debris: real traps, controlled sweeps                                  |
| PREREGISTRATION_CXR_DEVICE_TRAPS | chest-radiograph devices (CLiP) and controlled tube sweep                      |
| PREREGISTRATION_CXR_DRAIN        | real chest drains (in-ROI only)                                                |
| PREREGISTRATION_UNIVERSAL_I2E    | U-MtE, ablations U7–U14 (erase vs augment, protection, JTT, SPLICE, selectors) |
| PREREGISTRATION_SLAS             | patch-token artifact localisation and suppression                              |
| PREREGISTRATION_FINETUNE         | end-to-end fine-tuning (ResNet-50, ViT-S)                                      |
| PREREGISTRATION_INPAINT_LAMA     | LaMa inpainting baseline                                                       |
| PREREGISTRATION_NATURAL          | unaltered-distribution and leave-one-hospital-out tests                        |

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The primary family used for multiplicity control (`STATISTICAL_PLAN`) was assembled after the individual results were available; each member test was individually pre-registered, and the Holm correction is applied to the whole family.

## S2. Replication of the original pilot

Of 88 numbers reported in the thesis proposal, 73 are reproduced exactly (same sign, same confidence-interval verdict,  $|\Delta| \leq 0.02$ ), 12 agree in sign and significance with a larger difference, and 3 differ (a borderline occlusion interval, a contrast trap that was reported as null and is not, and one source stratum). Details: `results/SUMMARY.md`.

### S3. Datasets considered and not used

CANDID-PTX (not accessible outside Health New Zealand); RANZCR-CLiP devices linked to NIH (masking helped at both locations; device placement confounded with other devices); chest-radiograph cardiomegaly with lines over the heart (210 labelled images); TCGA/GrandQC pathology (pen-mark masks unreliable); ISIC ink (70 in-lesion images); breast ultrasound BUSI/BUS-BRA (caliper-free group only  $\approx 70\%$  pure after a two-detector design); DDTI thyroid (TI-RADS instead of pathology); BKAI-IGH colonoscopy specular highlights (no highlight-free images).

### S4. Artifact-mask audit and sensitivity

Visual audit of 40 random images per cell (thyroid detections: 95% on real calipers; caliper-free thyroid images: 95–97% without visible calipers; ovary: 40/40 detections are overlays, 31/40 calipers; caliper-free ovary images: 92% without calipers but  $\approx 28\%$  with other overlays). Capsule debris probe: pixel accuracy 0.92, IoU 0.62 against expert masks. Re-running with large-marker ( $\geq 50$  px) and strict-location ( $r \geq 0.7$  /  $r < 0.05$ ) definitions (thyroid, ovary) and a strict capsule Trap B leaves every primary conclusion unchanged (SENSITIVITY\_ARTIFACT\_LABELS).

### S5. Secondary method analyses

- *Patch-token suppression (SLAS)*. A linear probe on DINOv2 patch tokens localises real hair from 5 annotated images (patch AUROC 0.93; 0.97 with 50), but dropping flagged tokens improves reversed AUROC only modestly (+0.02 hair, +0.08 thyroid over ROI pooling).
- *Annotation-free pseudo-groups*. An overlay detector trained on synthetic insertion pairs recognises real calipers (AUROC 0.84) but not hair (0.57); pseudo-group balancing adds +0.04 on thyroid only.
- *Erasure combined with DFR*. Adds  $\leq 0.014$  over masking plus DFR.
- *Split-validation selector*. Admitting DFR as a candidate (trained and scored on disjoint validation halves) fixes the chest-drain case (0.656 vs 0.538) but makes choices noisier elsewhere; neither selector is uniformly best.

### S6. Reproducibility

All numbers in the main paper and this supplement are produced by scripts in the repository (`scripts/`); tables and figures are generated, never typed. A two-command tool applies U-MtE and SLAS to new data with any frozen backbone; it reproduces the experimental features exactly (cosine similarity 1.0).